

MD ANAWRUL KABIR FAHAD

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SUMMARY

Software Engineer with 1+ year of experience building a multi-tenant GPU platform on bare-metal Kubernetes and AWS at Poridhi.io, working on GPU virtualization, session orchestration, and backend services. Final-year Mechanical Engineering student at CUET.

EXPERIENCE

Poridhi.io

Junior Software Engineer

Remote, Bangladesh

Jun 2025 – Present

AI Studio: on-demand multi-tenant GPU platform (ai.poridhi.io)

- Cut GPU session launch time by ~70% (10–15 min → 3–5 min) by co-engineering the platform's migration to a bare-metal Kubernetes cluster with HAMi GPU virtualization, partitioning each RTX 4090 into 6×8 GB VRAM slices with hard isolation that Kubernetes time-slicing could not provide.
- Architected the initial AWS stack for on-demand GPU sessions: g4dn EC2 instances with custom AMIs, ECR-hosted TensorFlow images, JuiceFS-on-S3 session persistence, and sidecar containers for runtime isolation.
- Designed the PostgreSQL schema and REST API governing GPU slice allocation and session lifecycle, guaranteeing safe state transitions across concurrent Temporal workflows.
- Operated production cluster services (PostgreSQL, Temporal, Headlamp monitoring) and enforced least-privilege access with per-microservice Kubernetes RBAC.

TensorCode: in-browser GPU IDE (tensorcode.poridhi.io)

- Built the Hono.js backend on Cloudflare's edge stack (D1 storage, R2 file handling, KV session state) powering in-browser PyTorch and CUDA execution.
- Developed React frontend components and authored the team's architecture and deployment documentation, serving as the onboarding reference for new engineers.

Intern Software Engineer

May 2025 – Jun 2025

- Prototyped an authentication service (Flask, PostgreSQL, multi-arch Docker) and a Pulumi IaC workflow provisioning AWS EC2 instances and load balancers.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Go

Frameworks: React, Next.js, Node.js, Hono.js, Flask, PyTorch

Infrastructure & Cloud: Kubernetes, Docker, AWS (EC2, S3, ECR), Temporal, Terraform, Pulumi, Cloudflare, GitHub Actions

Databases: PostgreSQL, Redis, MongoDB, Cloudflare D1

EDUCATION

Chittagong University of Engineering & Technology (CUET)

B.Sc. in Mechanical Engineering

Chattogram, Bangladesh

Apr 2022 – Jul 2026 (expected)

PROJECTS

Parameter-Efficient Adaptation of a PINN Solver for High-Incidence Airfoil Flow | PyTorch

Ongoing thesis

- Developing a physics-informed neural network solver for flow over the NACA 0012 airfoil, using LoRA-style parameter-efficient fine-tuning to adapt it to out-of-distribution high-incidence conditions without full retraining. Supervisor: Prof. Dr. Md. Abu Mowazzem Hossain, CUET.

ACHIEVEMENTS

- **Champion**, Break The Monolith Hackathon (2025): microservices ride-sharing platform (Node.js, PostgreSQL, Redis, RabbitMQ).
- **Rising Team**, API Avenger Microservice Hackathon, CUET (2025): donation backend with idempotent webhooks and asynchronous event handling.
- **AI Engineering Hackathon, Brain Station 23**: built intent-based product search with a quantized mini-LLM, BERT, and a full observability stack.
- **Competitive programming**: ICPC Preliminaries (2022, 2023); CUET IUPC (2022–2023); 300+ problems solved across 50+ contests.